Gauss' theorem for UFDs

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Given a unique factorization domain R, and let us define primitive polynomials in $R[x]$ as all $p(x) = a_0 + a_1x + ...a_nx^n \in R[x]$ such that $\gcd_R(a_0, ..., a_n)$ is a unit in R.

Then, Gauss' lemma states that

1. (*Primitivity statement*) : The set of primitive polynomials in $R[x]$ is closed under multiplication.
2. (*Irreducibility statement*) : Given a non-constant polynomial $f(x) \in R[x]$,

   $f(x)$ is irreducible in $R[x] \Leftrightarrow f(x)$ is irreducible in $\text{Frac}(R)[x] \wedge f(x)$ is primitive in $R[x]$.

Given an infinite strictly increasing chain of principal ideals in $R[x]$, we can get an infinite strictly increasing chain of principal ideals generated by polynomials $p_1, p_2, ...$ all of which have the same degree. Suppose $p_i = a_i p_{i+1}$ for some a_i in R. None of a_i can be units in R. Now, the leading coefficients of $p_1, p_2, ...$ generate an ascending chain of principal ideals in R which is not possible as R is UFD. So, $R[x]$ satisfies ACCP.

If constant polynomial $p \in R[x]$ is irreducible, then it is also irreducible in R. Since R is UFD, so p is prime in R. So (p) is a prime ideal of $R[x]$. Consider the map $\hat{\pi} : R[x] \to (R/(p))[x]$ defined using $\pi : R \to R/(p)$ and $x \mapsto x$. Since (p) is prime ideal, so $R/(p)$ is domain, and so is $(R/(p))[x]$. Suppose p divides $f(x)g(x)$ for some $f(x), g(x) \in R[x]$. Then $\hat{\pi}(f(x)g(x)) = \hat{\pi}(f(x))\hat{\pi}(g(x)) = 0$, and since $(R/(p))[x]$ is integral domain, so assume WLOG that $\hat{\pi}(f(x)) = 0$, therefore, $f(x)$ is divisible by p in $R[x]$, showing that p is prime in $R[x]$.

If non-constant polynomial $f(x)$ is irreducible in $R[x]$, then by Gauss lemma, we have that $f(x)$ is irreducible in $\text{Frac}(R)[x]$ and primitive in $R[x]$. Suppose $f(x)$ divides $g(x)h(x)$ where $g(x), h(x) \in R[x]$. Since $\text{Frac}(R)$ is a field, therefore, $\text{Frac}(R)[x]$ is an Euclidean domain, and therefore a PID, and therefore an UFD, and therefore all irreducibles are prime in $\text{Frac}(R)[x]$. So $f(x)$ is prime in $\text{Frac}(R)[x]$, and WLOG assume that it divides $g(x)$ in $\text{Frac}(R)[x]$. Since $f(x)$ is primitive in $R[x]$, and it divides $g(x)$ (which belongs to $R[x]$) in $\text{Frac}(R)[x]$, so it also divides $g(x)$ in $R[x]$. So, $f(x)$ is also prime in $R[x]$.

Therefore, in $R[x]$, all irreducibles are prime.

Since $R[x]$ satisfies ACCP and all irreducibles are prime in $R[x]$, therefore, $R[x]$ is a UFD.

So finally, we get that whenever R is a UFD, so is $R[x]$.